

Power of Analogue Design

Analogue has become the new buzzword for design engineers who are on the lookout for efficient features when developing products that aim towards making life simpler and safer



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Today, life is full of electronics. Electronic products are everywhere around us. From the time we wake up in the morning to the time we sleep at night, we come across various simple and complex electronic devices and products.

A few decades ago, with the advent of computers and more so with personal computers, designers started focusing on digitising the world. Electronics was seen as a tool for digital revolution and became synonymous with the word 'digital.' Anything and everything that contained some kind of display was considered digital, while the real world was still dominated by analogue.

Interaction with the real world

happens through analogue, while the processing of data may be in digital. In general, humans experience the world analogically. Vision, for example, is an analogue experience where we perceive infinitely smooth gradations of shapes and colours. A foundational element in nearly every digital application, analogue chips convert real-world signals, such as voice, sound, pressure, temperature and electricity, into digital ones and zeros, manipulate and cleanse them, and deliver them as clearer, sharper and more precise signals that we rely on.

Resurrection

Of late, designers have re-discovered the beauty of analogue world and have started recognising the power of analogue circuits in their designs.

Analogue integrated circuits comprise one of the largest segments of the semiconductor industry. Analogue ICs are also vital parts of most of the circuit and product designs. For instance, for every embedded processor in a customer's electronic system, ten analogue components are typically required, although the count varies by system. And if an electronic device uses a battery or plugs into a wall, it will likely require at least one power management chip.

Most of the design engineers these days design with the key ideology of making the products simple and safe to use, so as to gain mass appeal for the product, thereby generating a rich user experience. These designers have recognised that the world is analogue, and even though they cannot escape processing the data and inputs digitally, they need to keep the input and output interfaces analogue.

Instances

A classic example would be in the field of medical test and measurement with the most common equipment being an electrocardiograph (ECG). The input to this equipment is always analogue, i.e., electrical signals being picked up by probes attached to a patient's body and the basic functions of an ECG machine including ECG waveform display, either through LCD screen or printed paper media, and heart rhythm indication as well as simple user interface through buttons.

The key challenge here for a designer is not how to process the data received from the probes, but more of signal acquisition due to the presence of the large DC offset and various interference signals. This potential can be up to 300 mV for a typical elec-